

Comparison of Different Optimization Techniques for Optimal Allocation of Multiple Distribution Generation

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Abstract – This paper aims to present and compare a number of optimization algorithms techniques namely, Whale Optimization Algorithm(WOA), Dragonfly Algorithm (DA) and Moth-Flame Optimization (MFO) algorithm. These approaches have been implemented to identify the optimal location and sizing of distributed generation (DG) in radial distribution systems considering minimization of network power losses. In this regard, multiple-DG units are simultaneously allocated and analyzed under two load power factors (i.e., unity and optimal). The proposed algorithms have been tested on the 69- and 119-bus radial distribution systems. The performance of the competitive algorithms has been compared with other evolutionary optimization methods under different system operating conditions in terms of all measures. The simulation results show the effectiveness of installing the proper size of DG at the suitable location based on different techniques.

Keywords - Optimization Algorithm, Radial Distribution system, Distributed generation, Power loss reduction.

I. INTRODUCTION

Distribution network is expanded in radial form, and the generation can't reach the existing demand So that , Distributed generation units are installed in the distribution network. DGs will affect the distribution system by changing The power flow of the distribution feeders,[1].

Distributed generation (DG) is an small electric power source has been connected directly to the load part of distribution system, [2,3]. DG can be classified into two categories .First category depend on energy sources of the DG include a)Nonrenewable Distributed Generation (NRDG) such as fuel cell (FC), gas turbine (GT) and micro-turbine .b) Renewable Distributed Generation (RDG) ex wind turbine and solar cell and , biomass . Nowadays ,14% of the total global energy demand is supplied by renewable energy sources .The integration of DG on the distribution system depends on the operation conditions of the system and the characteristics of the distributed generation,[4].Second category depend on the injected power ,include a)DG inject both Real power and reactive power such as Synchronous generator .b)DG inject Real power but consuming reactive power ex Wind turbine . c)DG inject only Real power such as PV, MT and FC with PE interface. d)DG inject Reactive power only such as Synchronous compensator, [5].

The integration of DG units into the distribution network affect the power flow .Different optimization techniques have been applied to solve the power flow problem. Optimization techniques are applied to determine the optimal allocation and the size of the DG to improve the performance of the distribution system .Optimization techniques have become more popular in recent years , and extend to include a different type of study .

A critical review of large number of literature related to the use of different algorithm implemented for optimal allocation and size of DG. Imperialist Competitive algorithm (ICA) and Genetic algorithm (GA) are used for power loss reduction, improving system voltage profile, increasing voltage stability index, load balancing [6]. Loss Sensitivity Factor (LSF) and the Bacterial Foraging Optimization Algorithm (BFOA) are used for minimizing power loss, operational costs and voltage profile[7]. Novel, Combined loss sensitivity, Index vector, and Voltage sensitivity index (VSI)methods used for reduction in energy losses, and reliability improvement[8] .Grey wolf optimizer (GWO) is used to minimized load buses voltage deviations and system power losses[9].novel Moth Swarm Algorithm (MSA), used for solve constrained Optimal Power Flow (OPF) problem[10]. Modified Grey Wolf Optimizer(MOGWO)has been applied to minimize the emission grades of deleterious pollutants and to reduce the generating cost individually[11].

This paper focused on the comparison of different optimization technique ,namely Whale Optimization Algorithm(WOA), Dragonfly Algorithm (DA) and Moth-Flame Optimization (MFO) algorithm) to determine the optimal allocation and size of the distributed generation (DG) in the radial distribution systems that consider reducing network power losses.

This paper is organized as follows: Section II discusses the formulation of the problem includes objective functions and constrains. Three optimization algorithms are explained in section III. The cases study and simulation results were discussed in section IV . Finally, the paper concludes in Section V.

II. PROBLEM FORMULATION

The proposed method aims to give the best location and size of new energy sources considering minimization of

network power losses. In this section the objective functions and constraints are explained.

A. Objective functions

Minimization Of Power Losses :Minimizing the total electrical power losses of distribution network is an important goal of integrating sources, which can be formulated as follows ,[12]:

$$P_{loss} = \sum_{i=1}^{nbr} R_i \quad (1)$$

Where R_i is resistance of branch i , I_i is current of branch I , nbr is number of branches in electric distribution network .

B. Constraints

The optimal allocation of DG in distribution system is subjected to the following constraints:

Generation constraints: For stable operation, real power outputs and reactive power outputs are restricted by the lower and upper limits as follows [7] :

$$P_{Gi}^{min} \leq P_{Gi} \leq P_{Gi}^{max} \quad (2)$$

$$Q_{Gi}^{min} \leq Q_{Gi} \leq Q_{Gi}^{max} \quad (3)$$

where P_{Gi} , Q_{Gi} are active and reactive generated power of bus i , P_{Gi}^{min} , Q_{Gi}^{min} is minimum active and reactive generated power of bus i and P_{Gi}^{max} , Q_{Gi}^{max} is maximum active and reactive generated power of bus i

Voltage Constraint :the constraints of load buses voltage magnitudes to be restricted within their limits as follows[7]:

$$V_i^{min} \leq V_i \leq V_i^{max} \quad (4)$$

where V_i is voltage of bus i , V_i^{min} is minimum voltage of bus i and V_i^{max} is maximum voltage of bus i

III. PROPOSED ALGORITHMS

The mathematical model for each proposed optimization algorithm (Whale Optimization Algorithm(WOA), Dragonfly Algorithm (DA) and Moth-Flame Optimization (MFO) algorithm) will be explained in this section.

A. Whale Optimization Algorithm

The special hunting behavior of humpback whales is described in this Algorithm ,this behavior is called bubble-net feeding method . The whales followed the typical bubbles that caused the '9-shaped path' while encircling prey during hunting. Simply bubble-net feeding/hunting behavior can be understood as the humpback whale went fell in water approximate 10-15 meter begin to produce bubbles in the spiral shape encircles prey and then follow the bubbles and moves upward the surface ,[13].

Mathematical model of WOA are as follow;

Encircling prey: Grey wolves encircle prey during the hunt per the following equations:

$$D = |C \cdot X_p(t) - X(t)| \quad (5)$$

$$X(t+1) = X_p(t) - A \cdot D \quad (6)$$

For t -th iteration, X_p and X are the victim and the grey wolf positions respectively. The coefficient vectors A and C can be calculated as follows:

$$A = 2a \cdot r_1 - a \quad (7)$$

$$C = 2 \cdot r_2 \quad (8)$$

where a decreased linearly from 2 to 0 and r_1 , r_2 are distributed randomly within the range $[0, 1]$.

Hunting: mimic the behavior of hunting process , We assume that there is a 50% probability of choice between either a shrinking encircle mechanism or a helical model to update the whale position during optimization The mathematical model is as follows:

$$X(t+1) = \begin{cases} X_p(t) - A \cdot D & \text{if } p < 0.5 \\ D' \cdot e^{bl} \cdot \cos(2\pi l) + X_p(t) & \text{if } p \geq .05 \end{cases} \quad (9)$$

$$D' = |X_p(t) - X(t)| \quad (11)$$

where p is a random number in $[0,1]$, b is a constant for defining the shape of the logarithmic spiral, l is a random number in $[-1,1]$.

step(1)read system data(bus and branch data) and define the upper and lower bounds of the variables(location and size of DG and bus voltage)

Step(2) Start algorithm .

Step(3) generate The initial population for each whales(a, A, C, l, p) randomly.

Step(4) Calculate fitness of objective function for each search agent(P_{loss}^* , a , A , C , l , and p).

Step(5) If $p < 0.5$ and $A < 1$, Update the leader by the Eq. 5.

Step(6) If $p < 0.5$ and $A \geq 1$, Select a random search agent and Update the position of the current search agent by the Eq. 6.

Step(7) if any search agent goes beyond the search space and amend it , Update P_{loss}^* .

Step(8) searching the better solution about minimized power loss and Check the stop criterion if it is satisfied then go to step(11).

Step(9) iter updating: Update the iter counter $iter = iter + 1$.

Step(10) New population: Create a new population of sitting of DG go to step (3) .

Step(11) Check the stop criterion, if it is satisfied then stop, else go to step(9).

B. Moth-Flame Optimization Algorithm

Moth-Flame Optimization (MFO) algorithm is a nature inspired algorithm based on the nature of navigation of moths. At night ,Moths fly by a maintained fixed angle with respect to the moon, and this mechanism is very effective for long distance travel in a straight line. But for short distances when the light source is very close , The moths fly spirally around the light. The moth eventually converges towards the light. Moths and flames are both solutions. Moths are used for actual search agents, to get the best position for moths using flames [14].

Different steps involved in the proposed algorithm are as follows:

The set of moths represented in a matrix form

$$M = \begin{bmatrix} m_{1,1} & \dots & m_{1,d} \\ \dots & \dots & \dots \\ m_{n,1} & \dots & m_{n,d} \end{bmatrix} \quad (12)$$

Where, n is the number of moths and d is the number of variables (dimension).

storing the corresponding fitness values in an array as follows

$$OM = \begin{bmatrix} OM1 \\ \dots \\ OMn \end{bmatrix} \quad (13)$$

set matrix similar to the moth matrix is considered as follows

$$F = \begin{bmatrix} F_{1,1} & \dots & F_{1,d} \\ \vdots & \ddots & \vdots \\ F_{n,1} & \dots & F_{n,d} \end{bmatrix} \quad (14)$$

storing the corresponding fitness value in an array as follows

$$OF = \begin{bmatrix} OF1 \\ \dots \\ OFn \end{bmatrix} \quad (15)$$

approximates the global optimal of the optimization problems is defined as follows:

$$MFO = (I, P, T) \quad (16)$$

Where I is a function that generates a random population of moths and corresponding fitness value. The P function is the main function which show the moves of the moths around the search space. The F function returns true if the termination criterion is satisfied and false if the termination criterion is not satisfied.

$I: \Phi \rightarrow \{M, OM\}$, $P: M \rightarrow M$, $T: M \rightarrow \{true, false\}$

Step(1) read system data and define the upper and lower bounds of the variables.

Step(2) Initialize The position of each moth with respect to a flame.

Step(3) Calculate the fitness of moths

Step(4) if $Ite=1$, Sort the first population of moths and Update the flames. else, Sort the moths and Update the flames

Step(5) obtained the best solutions so far are considered as the flames and stored the result in F matrix

Step(6) update the positions of the moths with respect to the best flame.

Step(7) in the final steps of iterations, Check the stop criterion

C. Dragonfly Algorithm

DA is a randomized search optimization algorithm that replicates the flying movement of dragonfly swarm. The behaviors of the dragonfly swarm have been discussed in two modes, namely hunting and migration mode. In hunting mode, small groups are formed and flied repeatedly over an area to find their prey. In migration mode, a big group formed, which will travel in a long distance with one destination location. These unique behaviors of dragonfly swarm become an

inspiration from the technique of exploration and exploitation in DA. The swarm consists of N dragonflies as search agents. The location and step vector of a search agent is P_{loss} and ΔP_{loss} respectively. with $1 \leq i \leq N$, [15].

the behavior of swarms follows three primitive principle (Separation, Alignment and Cohesion).

The separation is calculated as follows

$$S_i = \sum_{j=1}^N X - X_j \quad (17)$$

where X is the position of the current individual, X_j shows the position j -th neighbouring individual, and N is the number of neighbouring individuals.

Alignment is calculated as follows:

$$A_i = \frac{\sum_{j=1}^N V_j}{N} \quad (18)$$

where V_j shows the velocity of j -the neighboring individual.

The cohesion is calculated as follows:

$$C_i = \frac{\sum_{j=1}^N X_j}{N} - X \quad (19)$$

Attraction towards a food source is calculated as follows:

$$F_i = X^+ - X \quad (20)$$

where X^+ is the position of the food source.

Distraction outwards an enemy is calculated as follows:

$$E_i = X^- + X \quad (21)$$

where X^- is the position of the enemy.

The step vector shows the direction of the movement of the dragonflies and defined as follows

$$\Delta X_{t+1} = (sS_i + aA_i + cC_i + fF_i + eE_i) + w\Delta x_t \quad (22)$$

where s shows the separation weight, a is the alignment weight, c indicates the cohesion weight, f is the food factor, e is the enemy factor, w is the inertia weight, and t is the iteration counter.

the position vectors are calculated as follows:

$$X_{t+1} = X_t + \Delta X_{t+1} \quad (23)$$

Explanation the steps of the algorithm as follows:

Step(1) read system data and define the upper and lower bounds of the variables

Step(2) Initialize the dragonflies population X_t ($t = 1, 2, \dots, n$) Initialize step vectors ΔX_t ($t = 1, 2, \dots, n$).

Step(3) Calculate the objective values of all dragonflies (P_{loss})

Step(4) Update the food source and enemy, w , s , a , c , f , and e

Step(5) Update w , s , a , c , f , and e .

Step(6) Calculate S , A , C , F , and E , by using Eqs. (17) to (21)

Step(7) Update neighbouring radius

Step(8) Update velocity vector using Eq. (22)

Step(9) Update position vector using Eq. (23)

Step(10) update ΔP loss i and P loss i .

Step(11) Check the stop criterion.

IV. SIMULATION RESULTS

This section demonstrate the effectiveness of the proposed methods (WOA,DA and MFO) on the power flow of the distribution system. The proposed algorithms have been implemented on the IEEE 69-bus and 119-bus radial distribution system. simulations have been performed on four different cases showing table I .The results have been compared with the other optimization algorithms for each of the four cases mentioned.

An analytical software tool has been developed in MATLAB to run load flow, and determine optimal location and size of DG ;for calculate minimum power loss.

TABLE I. CASE STUDIES

Case #	Operation mode of DG	System
Case 1	DG operating at unity power factor	IEEE 69-bus radial distribution system
Case 2	DG operating at optimal power factor	
Case 3	DG operating at unity power factor	IEEE 119-bus radial distribution system
Case 4	DG operating at optimal power factor	

A. IEEE 69-bus radial distribution system

The IEEE 69-bus radial distribution system consists of 69 buses and 68 branches. The MVA and kV base of the test system are 100 MVA and 12.66 kV. The total real and reactive power loss of this case is 225 kW and 102.2 kVA. The bus data and line data of this system are taken from [16]. In this distribution network, multiple-DG units are simultaneously allocated and analyzed under two case depend on load power factors (i.e., unity and optimal).

1) Case 1. System with multiple DG at unity power factor

In this case DG are operated at unity power factor .the total power losses after determine the optimal location ,and size are tabulated in Table II. this results are compared with existing method and are tabulated in Table III. It is observed that the proposed method significantly have been reduced network power losses .

Fig.1 show The convergence curve of different optimization technique considering minimization of network power losses. MFO algorithm provides better result of the objective function. So that it converge before WOA and DA algorithms.

The real power losses are depicted in Fig. 2. The total power losses will be reduced after installing DGs with suitable size and location.

TABLE II. PERFORMANCE ANALYSIS OF THE SYSTEM AFTER DG INSTALATION CASE 1

Methods	WOA	DA	MFO
DG Location	49	66	11
	18	14	61
	61	61	21
DG Size(MW)	0.84046	0.23147	0.54917
	0.53352	0.52993	1.7458
	1.8084	1.7632	0.35254
P_{loss} (MW)	0.07019	0.0711	0.06946
Q_{loss} (MVAR)	0.03225	0.03557	0.03497
Time	252	13.5	13.9
V_{min} (pu)	0.97988(65)	0.97964(65)	0.97987(65)

TABLE III. COMPRRRATION OF ALGORITHM RESULTS FOR TEST SYSTEM CASE 1

Methods	DG Location	DG Size(MW)	P_{loss} (MW)
PSO[14]	61	1.1998	0.0832
	63	0.7956	
	17	0.9925	
GA[14]	21	0.9297	0.089
	62	1.0752	
	64	0.9848	
GA/PSO[14]	63	0.8849	0.0811
	61	1.1926	
	21	0.9105	
LSFSA[5]	18	0.4204	0.0771
	60	1.3311	
	65	0.4298	
WOA	49	0.84046	0.07019
	18	0.53352	
	61	1.8084	
DA	66	0.23147	0.0711
	14	0.52993	
	61	1.7632	
MFO	11	0.54917	0.06946
	61	1.7458	
	21	0.35254	

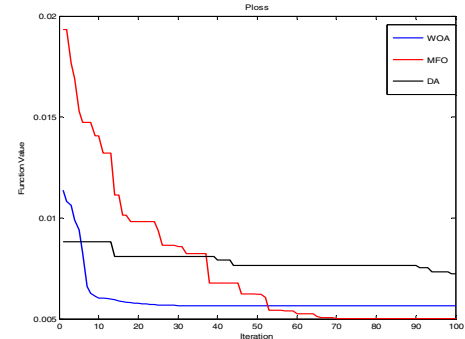


Fig 1. Performance of Optimization Algorithms (case 1)

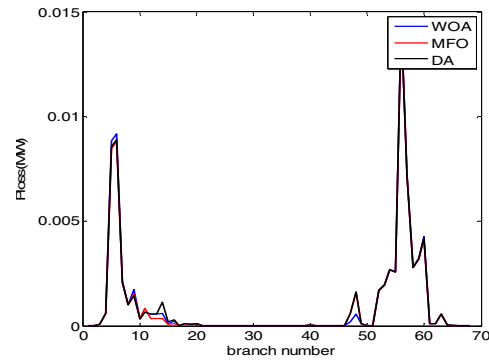


Fig 2. real power line loss (case1)

The voltage profile of the network has been improved after installing of DG,it is observed in Fig. 3.So,Minimum bus voltage is increased from 0.909202 p.u. with out DG, to 0.97988 p.u ,0.97964p.u and 0.97987p.u with DG .

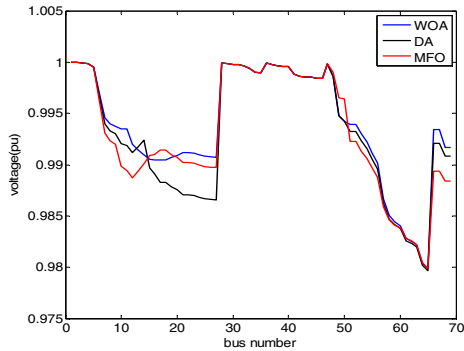


Fig 3. voltage profiles (case1)

2) Case 2 System with multiple DG at optimal power factor

In this case DG are operated at 0.85 lagging power factor. the total network power losses and minimum bus voltage after installing DG with suitable size and location are tabulated in Table IV. The results have been compared with other evolutionary optimization methods ,as in Table V.

TABLE IV. PERFORMANCE ANALYSIS OF THE SYSTEM AFTER DG INSTALATION CASE 2

Methods	WOA	DA	MFO
DG Location	61	11	11
	4	61	61
	17	17	18
DG Size(MVA)	1.8076	0.49634	0.51573
	1.9887	1.7481	1.7456
	0.53472	0.39501	0.38617
P _{loss} (MW)	0.00789	0.00501	0.005
Q _{loss} (MVAR)	0.00826	0.00708	0.00708
Time	235	21.7	13.2
V _{min} (pu)	0.99433(50)	0.99426(50)	0.99426(50)
V _{max} (pu)	1.0007	1.0008	1.0008

TABLE V. COMPRRATION OF ALGORITHM RESULTS FOR TEST SYSTEM CASE 2

Methods	Location	DG Size(MVA)	PF	P _{loss} (MW)
LSFSA[5]	18	.5498	0.866	0.01625
	60	1.1954	0.866	
	65	.3122	0.866	
GA/PSO [6]	18	362	0.71	0.0103
	61	1253	0.71	
	64	290	0.81	
GA/PSO [6]	11	549.5	0.85	0.00501
	21	351.9	0.85	
	61	1745	0.85	
WOA	61	1.8076	0.85	0.00789
	4	1.9887	0.85	
	17	0.53472	0.85	
DA	11	0.49634	0.85	0.00501
	61	1.7481	0.85	
	17	0.39501	0.85	
MFO	11	0.51573	0.85	0.005
	61	1.7456	0.85	
	18	0.38617	0.85	

The impact of DG integration on the network power losses and voltage profile of the system is shown in Fig.4 and 5, after the installation of the DG, the power losses in each branch are minimized and The voltage profile is also improved.

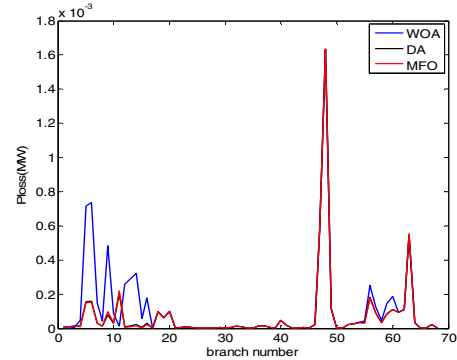


Fig 4. real power line loss (case 2)

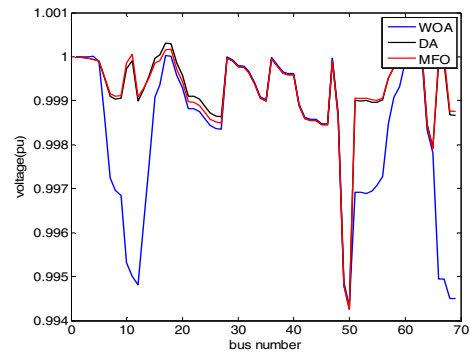


Fig 5. voltage profiles (case 2)

B. IEEE 119-bus radial distribution test system

The IEEE 119-bus radial distribution system consists of 119 buses and 118 branches, with a total real and reactive load of 22709.72 kW and 17041.07 kVar. The base MVA and base kV of the test system are 100 MVA (kV) and 11 kV, The bus data and line data of this system are taken from [18].

1) Case3. 119-bus system with DG at unity power factor

The total power loss after determine the optimal location ,and sizes of the DG operating at UPF are tabulated in table VI. The results are compared with existing method and are tabulated in Table VII.

TABLE VI. PERFORMANCE ANALYSIS OF THE SYSTEM AFTER DG INSTALATION CASE 3

Methods	WOA	DA	MFO
DG Location	39	110	50
	50	71	80
	79	80	109
	109	50	96
	73	41	72
DG Size(MW)	1.4809	2.8693	2.8833
	3.0047	2.8688	2.095
	2.6821	2.2825	3.1199
	3.1289	2.7676	1.6629
	2.7972	1.8867	2.5334
P _{loss} (MW)	0.59234	0.58185	0.57465
Q _{loss} (MVAR)	0.43720	0.43338	0.43240
Time	63.8	67.8	70.5
V _{min} (pu)	0.95727(54)	0.95424(54)	0.9541(54)

TABLE VII. COMPRATION OF ALGORITHM RESULTS FOR TEST SYSTEM CASE 3

Methods	DG Location	DG Size(MW)	P _{loss} (MW)	V _{min} (pu)
QOTLBO [7]	49	3.0135	0.5814	
	72	2.5435		
	82	1.6655		
	91	1.7662		
	109	3.317		
PFOA [7]	38	2.75	0.57897	0.9536
	46	2		
	71	2.8		
	91	2.16		
	109	2.95		
WOA	39	1.4809	0.59234	0.95727(54)
	50	3.0047		
	79	2.6821		
	109	3.1289		
	73	2.7972		
DA	110	2.8693	0.58185	0.95424(54)
	71	2.8688		
	80	2.2825		
	50	2.7676		
	41	1.8867		
MFO	50	2.8833	0.57465	0.9547 (54)
	80	2.095		
	109	3.1199		
	96	1.6629		
	72	2.5334		

Fig.6 show The convergence curve of different optimization technique considering minimization of network power losses. It is clear that, Superiority of MFO to access a global optimal solution.

Fig. 7, shows The effect of DGs installation on real power losses, It is clear that, the total power losses have been reduced.

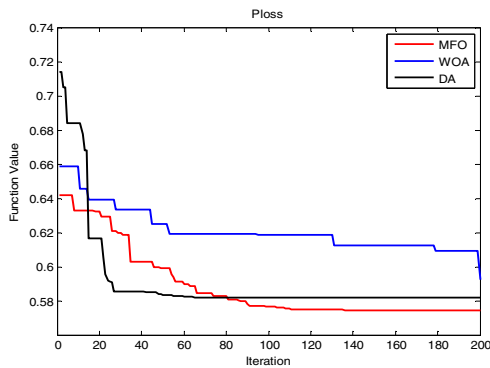


Fig 6. Performance of Optimization Algorithms (case 3)

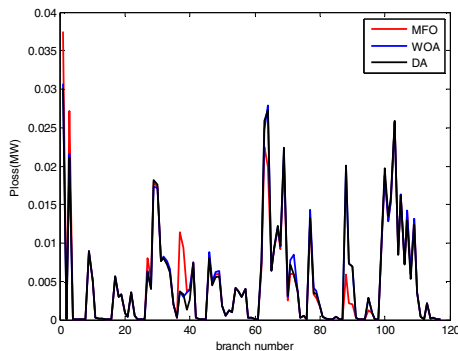


Fig 7. real power line loss (case 3)

It is observed from Fig. 8, That the voltage profile has improved after the installation of the DG operated at unity power factor

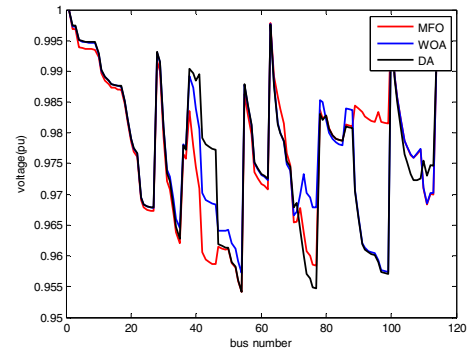


Fig 8. voltage profiles (case 3)

2) Case 4. 119-bus system with DG at optimal power factor

In this case DG are operated at 0.8 lagging power factor. The total power losses after determine the optimal location and size are tabulated in table VIII. The results are compared with existing method in Table IX. It is observed that, the proposed method significantly reduces the network power losses.

TABLE VIII. PERFORMANCE ANALYSIS OF THE SYSTEM AFTER DG INSTALLATION CASE 4

	WOA	DA	MFO
DG Location	79	2	110
	71	110	96
	21	4	72
	110	72	41
DG Size(MVA)	50	35	50
	2.9629	0	3.0989
	2.8219	3.1251	1.8892
	1.4867	3.4412	2.6296
	3.1834	2.875	2.027
	3.3348	3.2646	3.1009
P _{loss} (MW)	0.25681	0.35798	0.24161
Q _{loss} (MVAR)	0.16744	0.28134	0.18535
Time	9.35e+002	7.00e+001	6.50e+001
V _{min} (pu)	0.9612(46)	0.95937(99)	0.96543(85)

TABLE IX. COMPRATION OF ALGORITHM RESULTS FOR TEST SYSTEM CASE 4

Methods	Location	DG Size (MW)	P _{loss} (MW)	V _{min} (pu)
SA[5]	75	2.9296	0.684	0.93765 (54)
	116	1.5465		
	56	1.4841		
	36	4.4551		
	103	5.9126		
WOA	79	2.9629	0.25681	0.9612 (46)
	71	2.8219		
	21	1.4867		
	110	3.1834		
	50	3.3348		
DA	2	0	0.35798	0.95937 (99)
	110	3.1251		
	4	3.4412		
	72	2.875		
	35	3.2646		
MFO	110	3.0989	0.24161	0.96543 (85)
	96	1.8892		
	72	2.6296		
	41	2.027		
	50	3.1009		

Network power losses are minimized after the DG is installed at 0.8 lagging power factor, show Fig. 9.

Fig 10 depicts voltage profile of each bus in 119 bus distribution system after the installation of the DG operated at 0.8 lagging power factor .it is clear that ,the voltage profile has been improved

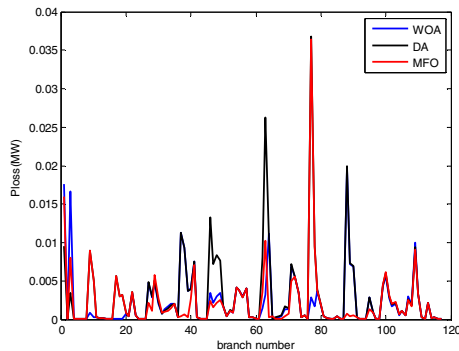


Fig 9. real power line loss (case 4)

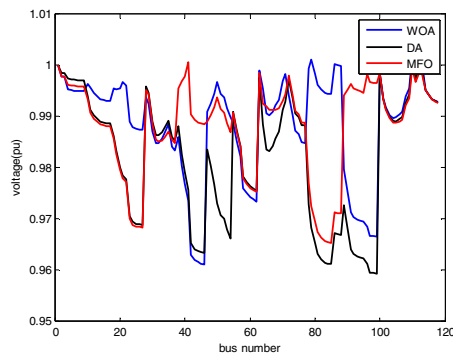


Fig 10. voltage profiles (case 4)

V. CONCLUSION

In this paper, a comparison of three optimization algorithms (WOA ,DA and MFO) have been proposed to identify the optimal location and sizing of distributed generation (DG) in radial distribution systems considering minimization of network power losses. The proposed methods have been tested on IEEE 69-bus and 119-bus radial distribution system .multiple-DG units are simultaneously allocated and analyzed under two load power factors (i.e., unity and optimal). It is observed from the simulation results that the DGs size obtained are higher at lagging power factor compared to the size obtained at unity power factor, however, the losses are found lower with DG at lagging power factor rather than DG at unity power factor, The voltage profile also improves with integration DG operating at lagging power factor than DG at unity power factor. MFO algorithm is better result than DA and WOA algorithms. The simulation result shows that the proposed methods provides better results when compared with other existing methods.

VI. REFERENCE

- [1] A. Saeed, K.Zare, and B.Mohammadi-Ivatloo. "Dynamic planning of distributed generation units in active distribution network." IET Generation, Transmission & Distribution , vol 9,no.12, pp .1455-1463,2015.
- [2] A.A.B.Rujula, "Future development of the electricity systems with distributed generation", Energy, vol. 34, no. 3, pp. 377–383,2009.

- [3] M.Kefayat., A. L. Ara, and S.A.N.Niaki. "A hybrid of ant colony optimization and artificial bee colony algorithm for probabilistic optimal placement and sizing of distributed energy resources." Energy Conversion and Management vol. 92,pp. 149-161,2015.
- [4] M. E .H. Golshan, and S. A. Arefifar. "Distributed generation, reactive sources and network-configuration planning for power and energy-loss reduction." IEE Proceedings-Generation, Transmission and Distribution vol.153,no .2 pp.127-136,2006.
- [5] I. S. Kumar, and N. P. Kumar. "A novel approach to identify optimal access point and capacity of multiple DGs in a small, medium and large scale radial distribution systems." International Journal of Electrical Power & Energy Systems vol. 45. No. 1 ,pp.142-151,2013.
- [6] M, H. Mohammad , Zeinalzadeh , A. Mohammadi, Y., and M.Abedini, "An efficient hybrid method for solving the optimal sitting and sizing problem of DG and shunt capacitor banks simultaneously based on imperialist competitive algorithm and genetic algorithm." International Journal of Electrical Power & Energy Systems .vol.54 , pp. 101-111,2014
- [7] K. R. Devabalaji, and K. Ravi. "Optimal size and siting of multiple DG and DSTATCOM in radial distribution system using bacterial foraging optimization algorithm." Ain Shams Engineering Journal vol.7, no.3, pp. 959-971, 2016.
- [8] M, V. V. S. N., and A.Kumar. "Comparison of optimal DG allocation methods in radial distribution systems based on sensitivity approaches." International Journal of Electrical Power & Energy Systems ,vol.53 ,pp.450-467,2013.
- [9] A. A. El-Gaafary et al., "Grey wolf optimization for multi input multi output system", Univers. J. Commun. Network, vol. 3, no. 1, pp. 1-6, 2015.
- [10]A. A. Mohamed et al., "Optimal power flow using moth swarm algorithm", Electr. Power Syst. Res., vol. 142, pp. 190-206, 2017.
- [11]A. A. Mohamed et al., "Multi-objective Modified Grey Wolf Optimizer for Optimal Power Flow", IEEE, 18th International Middle East Power Systems Conference (MEPCON), pp. 982-990, 2016.
- [12] A.R, M. , T. Niknam , A. Pahwa ,and A. K.Fard . "Multi-objective stochastic distribution feeder reconfiguration in systems with wind power generators and fuel cells using the point estimate method." IEEE Trans Power Syst; vol. 28,no. 2,pp.1483–1492,2013
- [13] S. Mirjalili and A. Lewis. "The Whale Optimization Algorithm." Advances in Engineering Software. vol. 4, pp.51-67,2016.
- [14] S.Mirjalili, . "Moth-flame optimization algorithm: A novel nature-inspired heuristic paradigm." Knowledge-Based Systems.vol.89,pp.228-249,2015
- [15] S.Mirjalili,,"Dragonfly algorithm: a new meta-heuristic optimization technique for solving single-objective, discrete, and multi-objective problems." Neural Computing and Applications vol. 27,no. 4 pp.1053-1073, 2016.
- [16] J.S.Savier, and D. Das. "Impact of network reconfiguration on loss allocation of radial distribution systems." IEEE Transactions on Power Delivery., vol. 22,no. 4 ,pp. 2473-2480,2007.
- [17] M. H. Moradi, and M. Abedini. "A combination of genetic algorithm and particle swarm optimization for optimal DG location and sizing in distribution systems." International Journal of Electrical Power & Energy Systems vol.34,no.1 ,pp.66-74,2012.
- [18] D .Zhang, Z. Fu, L .Zhang "An improved TS algorithm for loss minimum reconfiguration in large-scale distribution systems" Electric Power Syst Res vol. 77,no. 5,pp. 685–94, 2007.